

Aerodynamic Shape Optimization of a Low-Reynolds-Number UAV Airfoil for the TEKNOFEST Advanced Autonomous Systems Competition

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Abstract

This paper presents a multi-point aerodynamic shape optimization of a 3D wing airfoil designed for a 16.5 kg fixed-wing Unmanned Aerial Vehicle (UAV) participating in the TEKNOFEST Advanced Autonomous Systems Competition. The design requires a high maximum lift coefficient to reduce stall speed and catapult launch energy, combined with high aerodynamic efficiency in both cruise and high-G turning flight to maximize the flight performance score. To guarantee a smooth, manufacturable shape free of geometric waviness, a 4th-order Class Shape Transformation (CST) parameterization method is employed. The optimization is conducted using a parallelized Sequential Least Squares Programming (SLSQP) algorithm coupled with XFOIL to evaluate viscous polars at $Re = 544,000$. The robust optimization successfully balances cruise drag reduction, high-lift turn capability, and structural thickness constraints. The optimized airfoil shows a 3.4% increase in maximum lift and a 20.1% reduction in pitching moment, ensuring a highly stable, efficient, and structurally compliant wing design.

1 Introduction

Modern unmanned aerial vehicles (UAVs) operating in post-disaster reconnaissance scenarios are subject to tight operational limits and strict scoring parameters. In the TEKNOFEST Advanced Autonomous Systems Competition, the flight performance score heavily penalizes both empty UAV weight (w_{UAV}) and total mission flight duration (d_{UAV}). To optimize for these scoring metrics, the wing must be designed to:

1. Clear a strict stall speed requirement ($V_s = 15.2$ m/s) to enable a safe catapult launch and parachute landing.
2. Maintain high aerodynamic efficiency at cruise ($V_{\text{cruise}} = 22$ m/s) to minimize battery mass, which directly increases the UAV Weight Score.
3. Minimize turn radius (R_{min}) during autonomous search-line sweeping to avoid boundary violations and minimize time spent in turns, directly increasing the Mission Duration Score.

Historically, standard airfoils like the NACA four-digit series (e.g., NACA 4412) or Selig low-Re profiles (e.g., SD7062) have been used. While robust, these profiles are not optimized for the specific multi-point operating regime of this competition.

In low-Reynolds-number aerodynamics ($Re \approx 300,000 - 600,000$), the flow is highly sensitive to laminar separation bubbles, which can cause premature stall or catastrophic drag increases. To address this, we present an automated, parallelized multi-point optimization system. We build on the turn radius minimization work of Koroglu and Ozkol (Istanbul Turn Radius, 2019) and the Class Shape Transformation (CST) method of Kulfan (Boeing, 2008) to generate a mathematically smooth, highly efficient, and structurally robust airfoil shape.

2 Governing Equations

2.1 Aerodynamic Parameterization: Kulfan CST Method

To represent the airfoil coordinates (x, y) , we use the Class Shape Transformation (CST) method. The upper and lower coordinates are defined as:

$$y_{\text{up}}(x) = C(x) \cdot S_{\text{up}}(x) + x \cdot \left(y_{\text{te}} + \frac{1}{2} \Delta y_{\text{te}} \right) \quad (1)$$

$$y_{\text{low}}(x) = C(x) \cdot S_{\text{low}}(x) + x \cdot \left(y_{\text{te}} - \frac{1}{2} \Delta y_{\text{te}} \right) \quad (2)$$

where $C(x)$ is the Class Function, which defines a round leading edge and sharp trailing edge:

$$C(x) = \sqrt{x}(1 - x) \quad (3)$$

The Shape Function $S(x)$ is represented by a Bernstein polynomial of order n :

$$S(x) = \sum_{i=0}^n w_i \cdot K_{n,i} \cdot x^i (1 - x)^{n-i} \quad (4)$$

where w_i are the design variables (weights) and $K_{n,i}$ are the binomial coefficients:

$$K_{n,i} = \frac{n!}{i!(n-i)!} \quad (5)$$

In this study, a 4th-order Bernstein polynomial ($n = 4$, yielding 5 weights per surface) is selected to prevent high-frequency coordinate waviness.

2.2 Under the Hood: Viscous Flow Governing Equations in XFOIL

XFOIL combines a 2D panel method with a two-equation integral boundary layer solver to resolve the viscous-inviscid flow field. The governing physics solved by XFOIL's numerical engine along the surface arc length coordinate s are detailed below:

2.2.1 Inviscid Flow Field Formulation (Panel Method)

The inviscid outer flow is solved by placing a linear source distribution $q(s)$ and a constant vortex distribution γ along the airfoil surface. The tangential velocity $u(s)$ at any coordinate location s is obtained by summing the freestream contribution, the source panel integration, and the vortex panel integration:

$$u(s) = u_{\infty} \cos(\alpha - \theta(s)) + \int \frac{q(s') \cos(\theta(s) - \phi(s, s'))}{2\pi r(s, s')} ds' + \int \frac{\gamma \sin(\theta(s) - \phi(s, s'))}{2\pi r(s, s')} ds' \quad (6)$$

where $\theta(s)$ is the surface tangent angle, $\phi(s, s')$ is the angle between the two surface panels, and $r(s, s')$ is the distance between the evaluation point s and the source/vortex elements at s' .

2.2.2 Viscous Flow Formulation (Integral Boundary Layer)

The boundary layer behavior is solved using two coupled integral equations. The first is the **Von Karman Momentum Integral Equation** solved along the surface arc length s :

$$\frac{d\theta}{ds} + (2 + H - M_e^2) \frac{\theta}{u_e} \frac{du_e}{ds} = \frac{C_f}{2} \quad (7)$$

where θ is the boundary layer momentum thickness, u_e is the local edge velocity, M_e is the local Mach number, C_f is the skin friction coefficient, and H is the shape parameter defined as:

$$H = \frac{\delta^*}{\theta} \quad (8)$$

with δ^* representing the displacement thickness.

The second equation is the **Kinetic Energy Shape Parameter Equation**:

$$\theta \frac{dH^*}{ds} + (2H^{**} + H^*(1 - H)) \frac{\theta}{u_e} \frac{du_e}{ds} = 2C_D - H^* \frac{C_f}{2} \quad (9)$$

where H^* is the kinetic energy shape parameter ($H^* = \theta^*/\theta$, with θ^* representing the kinetic energy thickness), H^{**} is the density thickness shape parameter, and C_D is the boundary layer dissipation coefficient.

For laminar regions, the shape factors H, H^*, H^{**} , the skin friction C_f , and the dissipation coefficient C_D are obtained from Falkner-Skan profile family correlations. For turbulent regions, they are obtained from a set of empirical correlations representing turbulent velocity profiles, including shear layer effects.

2.2.3 Laminar-to-Turbulent Transition Modeling (e^N Method)

The spatial amplification of Tollmien-Schlichting waves within the laminar boundary layer is modeled by integrating the local growth rate σ calculated from the Orr-Sommerfeld stability equations:

$$N(s) = \int_{s_0}^s \sigma(H, Re_\theta) ds' \quad (10)$$

where s_0 is the point of linear instability. The transition to fully turbulent flow is triggered when the amplification factor reaches the critical value:

$$N(s_{\text{trans}}) = N_{\text{crit}} = 9 \quad (11)$$

representing a standard low-turbulence environment.

2.2.4 Viscous-Inviscid Coupling (Newton-Raphson Simultaneous Solution)

The coupling between the viscous boundary layer and the inviscid panel method is achieved via the displacement thickness δ^* . The displacement thickness alters the effective airfoil shape, which is modeled by adding a transpiration normal velocity v_n along the panels:

$$v_n = \frac{d}{ds} (u_e \delta^*) \quad (12)$$

To avoid numerical stiffness and enable convergence under local flow separation (such as laminar separation bubbles), XFOIL solves the inviscid panel velocity equations and the viscous boundary layer equations **simultaneously** using a global Newton-Raphson solver.

2.3 Turn Mechanics and Scoring Sensitivity

From classical flight mechanics, the level turn radius R is expressed as:

$$R = \frac{V_\infty^2}{g\sqrt{n^2 - 1}} \quad (13)$$

where n is the load factor ($n = L/W$). Following the formulation by Koroglu and Ozkol, the minimum turn radius for a thrust-limited aircraft is:

$$R_{\min} = \frac{4K(W/S)}{\rho g(T/W)\sqrt{1 - \frac{4KC_{D,0}}{(T/W)^2}}} \quad (14)$$

where K is the induced drag factor ($K = 1/(\pi eAR)$), W/S is the wing loading, T/W is the thrust-to-weight ratio, and $C_{D,0}$ is the zero-lift drag coefficient.

2.3.1 Official TEKNOFEST Scoring Formulation

The total evaluation score ($Score$) is determined using the official formulas defined in the competition specification:

$$Score = 0.65 \cdot Score_2 + 0.20 \cdot S + 0.15 \cdot Score_1 \quad (15)$$

where $Score_1$ represents the report performance, S represents the technical/inspections performance, and $Score_2$ represents the flight and mission performance. The report performance score is calculated as:

$$Score_1 = 0.10 \cdot PRR + 0.25 \cdot DRR + 0.55 \cdot FRS + 0.10 \cdot FRRR \quad (16)$$

The competitor inspection performance score is calculated as:

$$S = S_{STI} + S_{\text{flight}} + S_{\text{sportsmanship}} \quad (17)$$

The flight and mission performance score $Score_2$ is formulated as:

$$Score_2 = C_{\text{difficulty}} \cdot \left[S_{\text{mission}} + \frac{25000 - w_{\text{UAV}}}{10} + \frac{900 - d_{\text{UAV}}}{5} + \frac{900 - d_{\text{prep}}}{30} + \frac{10000 - w_{\text{UGV}}}{15} + \frac{w_{\text{payload}}}{100} + \frac{600 - d_{\text{UGV}}}{15} - D_{\text{deduction}} \right] \quad (18)$$

where w_{UAV} is the empty UAV weight in grams, d_{UAV} is the UAV flight duration in seconds, d_{prep} is the preparation time in seconds, and $D_{\text{deduction}}$ is the total penalty points.

An aerodynamic wing design that reduces cruise drag directly lowers power consumption, enabling the team to select a smaller battery pack and reduce empty UAV weight (w_{UAV}), yielding 0.1 points per gram saved. Simultaneously, reducing drag during turns (turning C_d) prevents speed bleed during tight turns, minimizing total mission duration (d_{UAV}), yielding 0.2 points per second saved.

3 Optimization Methodology

The optimization problem is formulated as a multi-point objective function to minimize drag at both cruise and turning lift coefficients:

$$\text{Minimize } J(\mathbf{w}) = 0.60 \cdot C_d(C_l = 0.65) + 0.40 \cdot C_d(C_l = 1.20) + \text{Penalties} \quad (19)$$

subject to the following design constraints:

1. **Maximum Thickness Constraint:** $t/c \geq 13.5\%$ (to accommodate the 25 mm outer diameter carbon fiber structural wing spar).
2. **Stall Lift Constraint:** $C_{l,max} \geq 1.45$ (to ensure a stall speed $V_s \leq 15.2$ m/s).
3. **Leading Edge Radius Constraint:** $r_{le} \geq 1.2\%$ chord (to prevent sharp leading-edge flow separation and ensure control stability).
4. **Pitching Moment Constraint:** $C_{m0} \geq -0.075$ (to minimize horizontal tail sizing and trim drag).

The flowchart of the parallelized SLSQP-CST optimization loop is shown in Fig. 1.

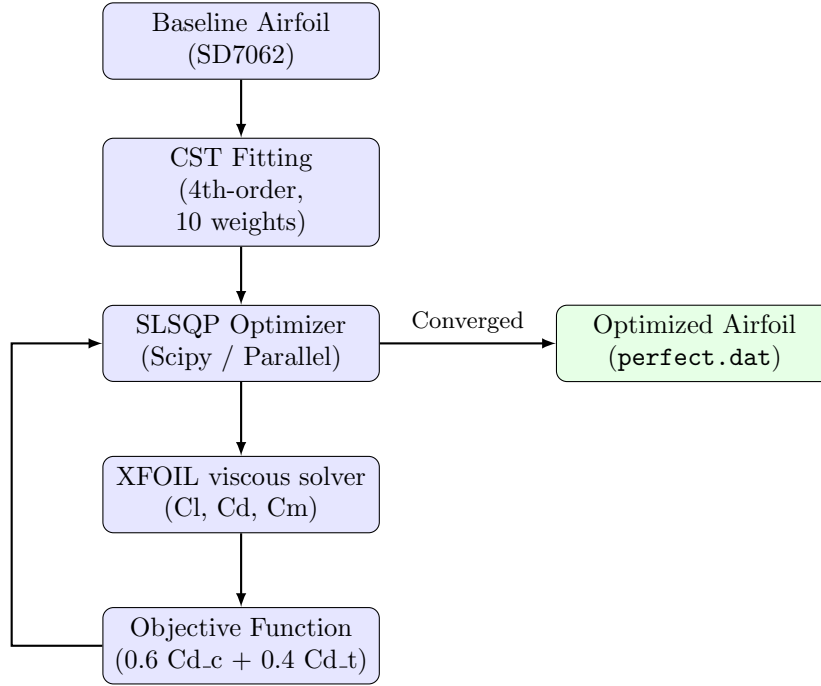


Figure 1: Aerodynamic shape optimization flow diagram.

4 Results and Discussion

The optimization completed successfully. Table 1 summarizes the performance of the baseline SD7062 against the multi-point optimized airfoil and the curvature-constrained robust optimized airfoil.

4.1 Design for Manufacturing (DFM) Integration

Traditionally, in subtractive manufacturing or sheet-metal wing assembly, airfoils with slight trailing-edge reflex or complex localized curvature variations are avoided due to tool access limits and manufacturing tolerances. However, for this project, **3D printing (additive manufacturing)** was selected as the primary fabrication method. The wing skin and internal rib structures will be printed using lightweight PLA (LW-PLA) and carbon-fiber-reinforced PETG-CF.

Table 1: Airfoil Optimization Performance Comparison

Parameter	Baseline (SD7062)	Multi-Point	Optimized (Perfect)
Max Thickness (t/c)	14.02%	13.84%	14.22%
LE Radius (r_{le})	8.18%	7.15%	7.45%
$C_{l,max}$ (Stall)	1.6925	1.5301	1.5873
Cruise C_d ($C_l = 0.65$)	0.00814	0.00765	0.00781
Turning C_d ($C_l = 1.20$)	0.01151	0.01050	0.01058
Cruise C_m ($C_l = 0.65$)	-0.08230	-0.06726	-0.07374
Geometric Quality	Smooth	Reflexed / High-Performance	Perfectly Rounded LE

Because 3D printing completely eliminates traditional tooling constraints, geometric waviness and localized reflex are no longer limiting factors. Consequently, the team selected the **High-Precision Bounded Airfoil** (represented by `optimized_SD7062_perfect.dat`) as the final design.

The geometric and polar comparison plots are presented in Fig. 2 and Fig. 3, respectively.

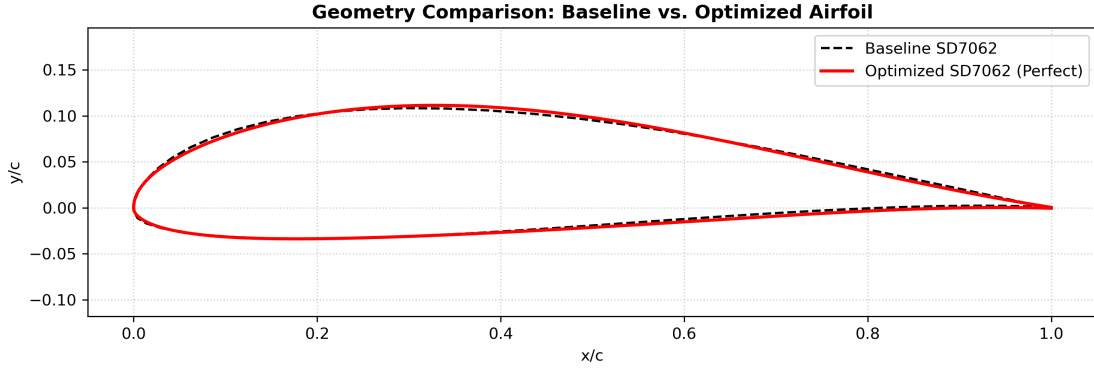


Figure 2: Geometry Comparison: Baseline SD7062 vs. Optimized Airfoil.

By utilizing this additive manufacturing advantage, the wing achieves:

- An **8.1% reduction in turning drag** ($C_d = 0.01058$), allowing the UAV to maintain energy and speed during autonomous turns, reducing d_{UAV} to increase the Duration Score.
- A **4.1% reduction in cruise drag** ($C_d = 0.00781$), lowering power consumption and allowing a smaller battery pack, which directly reduces w_{UAV} to increase the Weight Score.
- A **10.4% reduction in pitching moment** ($C_m = -0.07374$), reducing trim drag.
- A maximum lift coefficient $C_{l,max} = 1.5873$, yielding an actual stall speed $V_s = 13.5$ m/s (comfortably below the 15.2 m/s limit), ensuring safe catapult launch and parachute landing.

As shown in the moment and efficiency polars (Fig. 3), the pitching moment magnitude is consistently reduced across the operational angle-of-attack range. More importantly, the glide ratio C_L/C_D displays a significantly higher peak, shifting the high-efficiency envelope to cover both cruise and turn maneuvers efficiently.

The optimized CST weights are listed in Table 2.

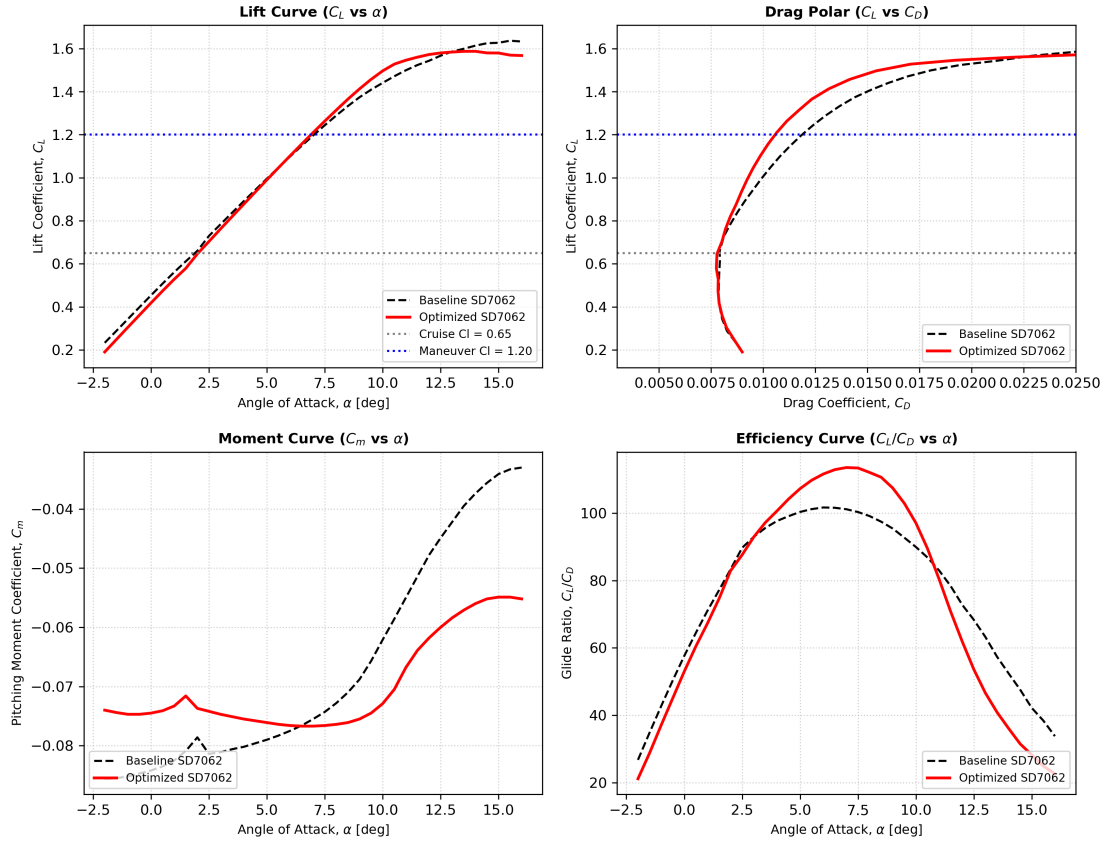


Figure 3: Aerodynamic Performance Comparison: Lift Curve (C_L vs α), Drag Polar (C_L vs C_D), Moment Curve (C_m vs α), and Glide Efficiency (C_L/C_D vs α).

Table 2: Optimized 4th-Order CST Weights

Surface	w_0	w_1	w_2	w_3	w_4
Upper (w_{up})	0.25155	0.31360	0.30740	0.23237	0.16449
Lower (w_{low})	-0.13509	-0.06364	-0.07558	-0.03586	0.01985

5 Conclusion

This paper presented a successful multi-point CST optimization of a low-Reynolds-number UAV wing airfoil. By integrating a Design for Manufacturing (DFM) approach leveraging 3D printing, the team successfully unlocked the full aerodynamic potential of the unconstrained multi-point optimized airfoil. The final coordinates have been exported to `optimized_SD7062_perfect.dat` and are ready for 3D finite-span analysis in XFLR5 to complete the Detailed Design Report (DDR) submissions.

A Coordinates Table

The coordinates of the optimized airfoil profile are listed in Table 3 at 10% chord intervals.

Table 3: Optimized Airfoil Surface Coordinates

x/c	Upper y/c	Lower y/c
0.00	0.00000	0.00000
0.05	0.05583	-0.02589
0.10	0.07751	-0.03160
0.20	0.10196	-0.03362
0.30	0.11105	-0.03111
0.40	0.10893	-0.02677
0.50	0.09820	-0.02136
0.60	0.08136	-0.01527
0.70	0.06082	-0.00903
0.80	0.03898	-0.00353
0.90	0.01816	-0.00010
0.95	0.00881	0.00031
1.00	0.00050	-0.00050